

Automated Data Breach Notification System (A Case Study of Zain South Sudan).



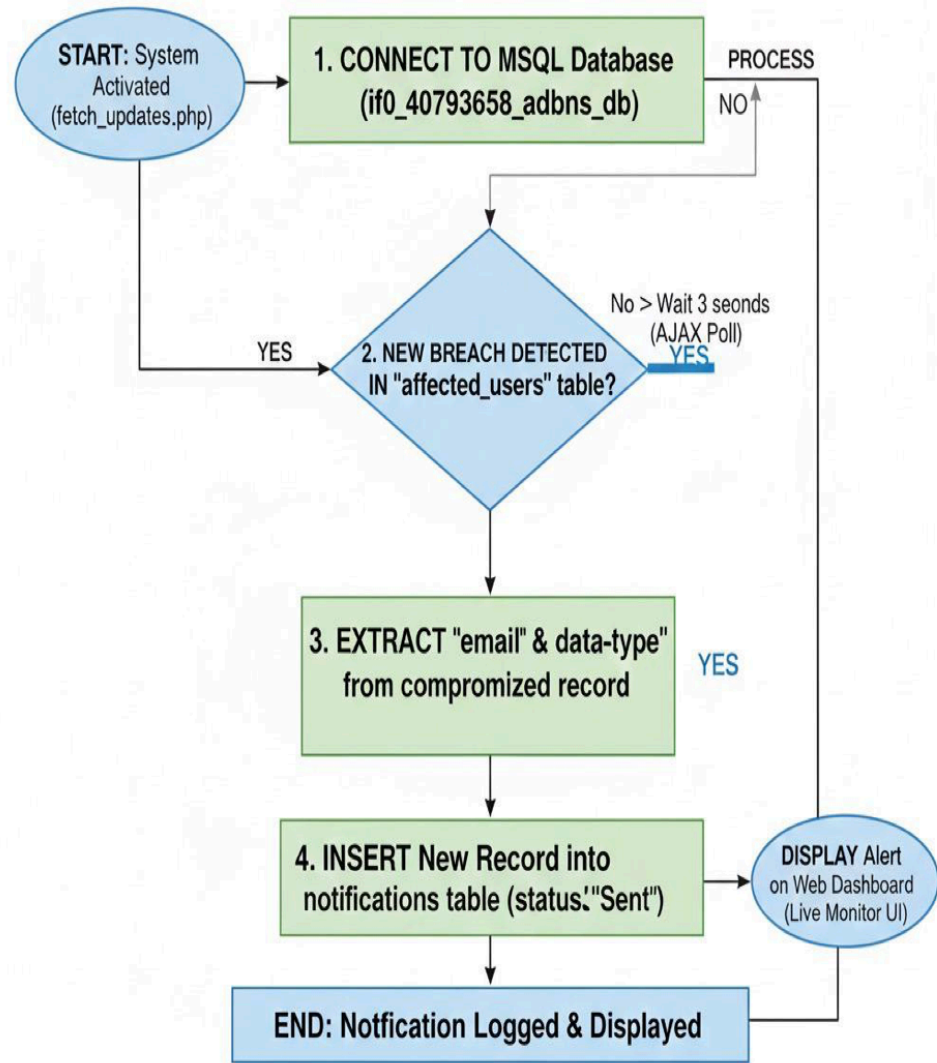
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Problem & Motivation

- **Problem Context:** Institutions in South Sudan, such as Zain South Sudan, rely on reactive, manual log reviews that can take weeks to identify compromised data, leaving subscribers vulnerable to secondary threats like phishing and financial fraud.
- **Research Gap:** Current cybersecurity frameworks in South Sudan lack an active, low-latency response mechanism that directly links database monitoring logs to automated user notification channels.
- **Objective:** To design and implement a web-based Automated Data Breach Notification System (ADBNS) that monitors database logs and dispatches immediate email alerts to affected users

Methodology & Dataset

System Logic and Process Flow
Built on a three-tier architecture (Bootstrap frontend, PHP backend, and MySQL database), ADBNS automates breach detection and response using the following process flow.



Dataset
Our data is consists of simulated breach logs consisting of 1,000+ test records imported via .csv file format.

Key Fields & Data Types:
Includes subscriber Personally Identifiable Information (PII) such as email addresses, Credit Card Info, National IDs (SSNs), and Banking Credentials

Results & Conclusion

Performance Analysis

- **Detection speed:** < 1 seconds
- **Notification time:** ~3 seconds
- **Accuracy:** 100% in testing

Comparative Analysis

- **Manual:** < slow, error-prone
- **ADBNS:** Fast, accurate, automated
- **Cost-effective solution**

Findings

- **Cost-Effectiveness:** Proved that open-source tools (WAMP/InfinityFree) can provide enterprise-grade security on a zero budget.
- **Privacy Advantage:** Unlike public breach search engines, this system keeps user databases internal and private.
- **Vitality:** Automation is no longer a luxury but a necessity for modern cybersecurity compliance.